Jessica M. **Caleta**

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Computational molecular epidemiology; pathogen evolution and cross-species transmission; viral genomic surveillance; Bayesian phylodynamics; wastewater-based epidemiology; AMR surveillance; reproducible bioinformatics methods for public health.

Education

Sep 2017 - Dec 2020 — **Master of Science, Microbiology and Immunology, University of British Columbia**

- **Thesis:** Quantification of intrahost genetic diversity of *Piscine orthoreovirus* (PRV-1) infecting Atlantic and Chinook salmon.

Sep 2013 - Apr 2017 — **Bachelor of Science with Distinction, Combined Major in Microbiology and Oceanography, University of British Columbia**

- Completed Science One, a selective first-year interdisciplinary science program.

Research experience

Oct 2026 - present — **Research Affiliate, Brown University | Volunteer | Dr. Brenda Rubenstein**

- Investigating dynamic protein–protein interfaces using clinically derived protein–protein interaction (PPI) networks across cancer subtypes.

Apr 2025 - Mar 2026 — **Genomics Specialist, Provincial Health Services Authority, British Columbia Centre for Disease Control**

- Enabled *Treponema pallidum* (syphilis) AMR surveillance in 16S/23S rRNA genes by building Nextflow workflow, riboscope, to detect mutations from paired-end sequencing of amplicon libraries.
- Optimized wastewater WGS spending through statistical analysis of sequencing cadence, cutting annual costs by 50%.

Jun 2022 - Mar 2025 — **Genomics Liaison Technical Officer, Public Health Agency of Canada, National Microbiology Laboratory**

- Established WGS analysis for BC's first wastewater surveillance program to monitor emerging SARS-CoV-2 variants at 15 sites across 5 Health Authorities, by developing and validating Nextflow pipeline WasteFlow, managing data in Linux HPC cluster, and delivering biweekly reports to National Laboratory and YVR International Airport.
- Identified origin of Canada's first domestic influenza A(H5N1) human case within 2 days using Bayesian evolutionary analysis, contributing to publications in *Emerging Infectious Diseases* and *The New England Journal of Medicine*.
- Accelerated more than 100 routine phylogenomic analyses threefold by automating data transfers, refactoring R code from dplyr to data.table, and migrating workflows to SLURM-enabled Nextflow pipelines, including fluflo and covflo.
- Developed epyflu, a Python package for programmatic retrieval and local SQLite tracking of international sequence-database records.

Jul 2021 - Jun 2022 — **Genomics Analyst, British Columbia Centre for Disease Control**

- Published mink sample type with best SARS-CoV-2 yield to conserve resources, through compiling dataset of 2000+ observations, statistical analysis, and visualization in R
- Developed BEAST-FLOW, a Nextflow workflow for Bayesian evolutionary molecular-clock analysis.
- Designed an R algorithm that identified 291 reinfections in British Columbia surveillance data.
- Built an R Shiny application with Bash scripts to automate secure data transfers between BCCDC and Canada's Michael Smith Genome Sciences Centre.

Sep 2017 - Dec 2020 | **MSc Researcher, Viral Genomics, University of British Columbia | Dr. Curtis Suttle**

- Investigated intrahost genetic diversity of piscine orthoreovirus-1 (PRV-1) in salmon, identifying a genetically distinct variant in wild Columbia River Chinook and a preliminary association between elevated within-host diversity and persistent heart and skeletal muscle inflammation (HSMI).
- Designed DiVirusity, a Python, Bash, and R pipeline that quantifies intrahost genetic diversity in segmented viral genomes directly from paired-end short reads; developed sequencing-project quality-control tools.
- Collaborated with researchers at UBC, Fisheries and Oceans Canada, and the Pacific Salmon Foundation; coauthored research on the global transmission of piscine orthoreovirus between farmed and wild salmon.

Jan 2017 - Aug 2017 | **Undergraduate Researcher, University of British Columbia | Dr. Curtis Suttle**

- Designed fluorescent molecular beacons targeting 16S rRNA to detect extracellular ribosomes in seawater as an indicator of prokaryotic cell lysis, refining probe sequences for taxonomic coverage and hairpin structure with research associate Dr. Kevin X. Zhong; summer research funded by NSERC Undergraduate Student Research Award.
- Developed a modified FISH-STIC protocol and optimized beacon hybridization conditions for flow-cytometric detection of ribosomes, testing purified E. coli ribosomes and seawater samples; continued the project into the first eight months of MSc.

Journal Publications

1. Griffiths, E. J., Cameron, R., Barclay, C., John, N. S., Dooley, D., Gill, I., Iseminger, M., Anwar, M. Z., Horsman, M. E., MacKenzie, K., Prystajecky, N., Tyson, J., Jassem, A., Lee, T., Azana, R., Fung, J., Chan, M., Cheung, B., Tsang, F., Fornika, D., **Caleta, J. M.**, Newman, T., Yang, K. C., Russell, S. L., Zlosnik, J. E. A., Hoang, L., Knox, N., Tyler, A. D., Haidl, E., Yadav, C., Duggan, A. T., Kearney, L., Townend, C., Cruz, B., Van Domselaar, G., & Hsiao, W. W. L. (2026). The mpox contextual data specification package: A data curation toolkit to support collaborative pathogen genomic surveillance. *Microbial Genomics, 12*(1), Article 001614. https://doi.org/10.1099/mgen.0.001614

2. Himsworth, C. G., **Caleta, J. M.**, Jassem, A. N., Yang, K. C., Zlosnik, J. E. A., Tyson, J. R., Wilson, L., Kuchinski, K. S., Giacinti, J., Willie, M., Redford, T. D., Winchester, M., Thacker, C., Berhane, Y., Burns, T., Prystajecky, N., & Russell, S. L. (2025). Highly pathogenic avian influenza A(H5N1) in wild birds and a human, British Columbia, Canada, 2024. *Emerging Infectious Diseases, 31*(6), 1216 - 1221. https://doi.org/10.3201/eid3106.241862

3. Jassem, A. N., Roberts, A., Tyson, J., Zlosnik, J. E. A., Russell, S. L., **Caleta, J. M.**, Eckbo, E. J., Gao, R., Chestley, T., Grant, J., Uyeki, T. M., Prystajecky, N. A., Himsworth, C. G., MacBain, E., Ranadheera, C., Li, L., Hoang, L. M. N., Bastien, N., & Goldfarb, D. M. (2025). Critical illness in an adolescent with influenza A(H5N1) virus infection [Letter to the editor]. *The New England Journal of Medicine, 392*(9), 927–929. https://doi.org/10.1056/nejmc2415890

4. Li, F. F., Faber, A., **Caleta, J. M.**, Goldfarb, D. M., Sekirov, I., Prystajecky, N. A., Srigley, J. A., Mishaal, R., & Jassem, A. N. (2025). Clinical application of phage immunoprecipitation sequencing to diagnose enterovirus D68 as the underlying etiology in a case of Guillain-Barré syndrome. *The Journal of Infectious Diseases, 231*(2), 490–494. https://doi.org/10.1093/infdis/jiae411

5. Andrew, C. L., Russell, S. L., Coombe, M., Zlosnik, J. E. A., Kuchinski, K. S., **Caleta, J.**, Fjell, C., Berhane, Y., Bowes, V., Redford, T., Thacker, C., Wilson, L., Henaff, M., Harms, N. J., Jassem, A., Giacinti, J., Soos, C., Prystajecky, N., & Himsworth, C. (2024). Descriptive epidemiology and phylodynamics of the “first wave” of an outbreak of highly pathogenic avian influenza (H5N1 clade 2.3.4.4b) in British Columbia and the Yukon, Canada, Apr to Sep 2022. *Transboundary and Emerging Diseases, 2024*(1), Article 2327939. https://doi.org/10.1155/2024/2327939

6. Himsworth, C. G., **Caleta, J. M.**, Coombe, M., McGregor, G., Dibernardo, A., Lindsay, R., Sekirov, I., & Prystajecky, N. (2023). A comparison of sampling and testing approaches for the surveillance of SARS-CoV-2 in farmed American mink. *Journal of Veterinary Diagnostic Investigation, 35*(5), 528–534. https://doi.org/10.1177/10406387231183685

7. Boyd, E., Coombe, M., Prystajecky, N., **Caleta, J. M.**, Sekirov, I., Tyson, J., & Himsworth, C. (2023). Hands off the mink! Using environmental sampling for SARS-CoV-2 surveillance in American mink. *International Journal of Environmental Research and Public Health, 20*(2), Article 1248. https://doi.org/10.3390/ijerph20021248

8. Mordecai, G. J., Miller, K. M., Bass, A. L., Bateman, A. W., Teffer, A. K., **Caleta, J. M.**, Di Cicco, E., Schulze, A. D., Kaukinen, K. H., Li, S., Tabata, A., Jones, B. R., Ming, T. J., & Joy, J. B. (2021). Aquaculture mediates global transmission of a viral pathogen to wild salmon. *Science Advances, 7*(22), Article eabe2592. https://doi.org/10.1126/sciadv.abe2592

9. Zhong, K. X., Suttle, C. A., Baudoux, A.-C., Derelle, E., Colombet, J., Cho, A., **Caleta, J.**, Six, C., & Jacquet, S. (2018). A new freshwater cyanosiphovirus harboring integrase. *Frontiers in Microbiology, 9*, Article 2204. https://doi.org/10.3389/fmicb.2018.02204

Invited talks & conference presentations

- **Caleta, J. M.**, & Newman, T. (2025, Feb). *Bioinflow* [Poster presentation]. National GLTO Summit, National Microbiology Laboratory, Winnipeg, Manitoba, Canada.
- **Caleta, J. M.** (2024, Jun). *The implementation of WGS surveillance in BC wastewater: "The making of a pipe dream"* [Invited talk]. Public Health Agency of Canada Wastewater Working Group meeting, virtual.
- **Caleta, J. M.** (2024, Feb). *The implementation of WGS surveillance in BC wastewater: "The making of a pipe dream"* [Invited talk]. National GLTO Summit, National Microbiology Laboratory, Winnipeg, Manitoba, Canada.
- **Caleta, J. M.** (2019, Feb). *Quasispecies dynamics of Piscine orthoreovirus (PRV) in BC salmon* [Oral presentation]. Pacific Ecology and Evolution Conference, Bamfield, British Columbia, Canada.

Teaching & mentoring

Apr 2025 - Jun 2025 | **Sessional Lecturer and Course Developer, University of Lethbridge**

- Designed and taught graduate course in Advanced Bioinformatics with colleague.
- Delivered 12 lectures, designed assignments and assessment criteria, held office hours, and supervised student research projects.

Sep 2018 - May 2019 | **Professor's Assistant, Marine Microbiology, University of British Columbia**

- Supported instruction and learning for more than 70 students in an upper-level marine microbiology course, providing guidance on over 200 exams and assignments.
- Organized additional sessions to explain course material and address student questions.

Oct 2018 - May 2019 | **Tutor, Oxford Learning Center**

- Taught up to three students at different grade levels simultaneously, adapting explanations and assigning and assessing homework to support individual learning needs.

May 2015 - Aug 2015 | **Tutor, Tutor Doctor**

- Tutored elementary and secondary students in chemistry and mathematics; developed individualized study plans and targeted practice activities.

Awards & research funding

- **Dr. Lucie Dutil Memorial Award in Applied Science and Research**, 2023, Public Health Agency of Canada (PHAC)
  Team recognition as a member of AGILE / the Canadian Public Health Laboratory Network (CPHLN) COVID Genomics Program (CCGP). Recognizes outstanding contributions to science and technology and the application of scientific information to public health protection and promotion.
- **Mitacs Accelerate Fellowship**, 2019, Mitacs and Pacific Salmon Foundation | CAD $30,000
  Funds scientists whose graduate research contributes new and relevant findings to their field.

- **Ocean Leaders Fellowship**, 2018, University of British Columbia | CAD $10,000
  Interdisciplinary training connecting marine research with policy and management.
- **Undergraduate Student Research Award**, 2017, Natural Sciences and Engineering Research Council (NSERC) | CAD $4,500
  Awarded to natural science students with strong academic record and research aptitude
- **All Canadian Award**, 2017, Canadian University Synchronized Swimming League
  Recognition of academic achievement (≥3.4GPA) while competing in university sport during 2015 - 2016.
- **Chancellor's Scholar Award**, 2013, University of British Columbia
  Recognizes incoming undergraduate students with outstanding academic achievement ≥95% (4.0GPA).

Technical expertise

- **Programming & analysis:** Python, R (dplyr, data.table), Bash, SQL/SQLite, statistical analysis, and data visualization.
- **Reproducible computing:** Nextflow workflow development, Git, Linux, high-performance computing, SLURM/SGE, Docker/Apptainer, mamba, and renv.
- **Pathogen genomics:** Whole-genome and amplicon sequencing analysis, viral intrahost diversity, Bayesian phylogenetics and molecular-clock analysis (BEAST), wastewater genomic surveillance, resistance-mutation detection, and genomic-data management.
- **Laboratory training:** Nucleic-acid extraction, PCR, gel electrophoresis, microbial culture, flow cytometry, and molecular-beacon design.

Leadership & community service

Aug 2025 - present	**ICC-Certified Skipper, Courageous Sailing Club, Boston** - Captained seven keelboats up to 37 feet, leading crews of 3–4 on day and overnight passages with responsibility for planning, navigation, weather assessment, and safety. Earned the International Certificate of Competency through IYT in 2025.
Oct 2021 - Oct 2023	**Social Media Manager, UBC Sailing Club** - As a Rear Commodore on the steering committee, doubled social media following for Canada's largest nonprofit sailing club in under a year, planned events, and designed marketing brochure.
May 2021 - Nov 2021	**Lessons Coordinator & Head of Communications, UBC Sailing Club** - Coordinated club lessons while managing member communications, produced newsletters for 1500+ readership, and introduced feedback initiatives.
Dec 2017	**Volunteer, Moria Refugee Camp, Greece** - Worked with an international team to distribute supplies, organize shelter, and make repairs; communicated across language barriers.
Nov 2016 - Apr 2017	**Research Volunteer, BC Children's Hospital** - Organized and interpreted data supporting a national study of Crohn's disease and ulcerative colitis.
Jun 2011 - Aug 2014	**Summer Camp Counselor, Camp Latona** - Led outdoor activities for more than 50 children, managed activity safety, resolved conflicts, and promoted inclusion and teamwork.